



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



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di Fisica
e Astronomia
Galileo Galilei

No U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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No U-Turn Sampler

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However, it requires the tuning of two user-specified parameters:
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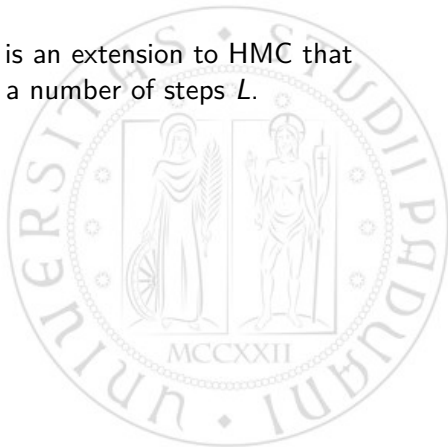
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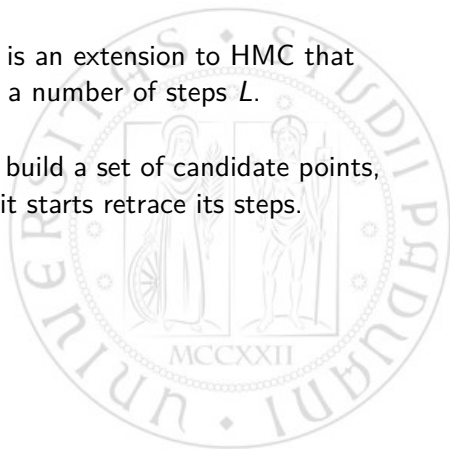
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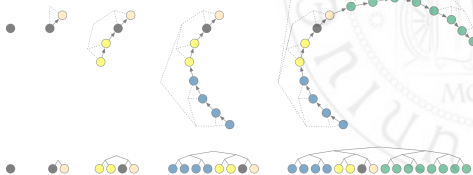
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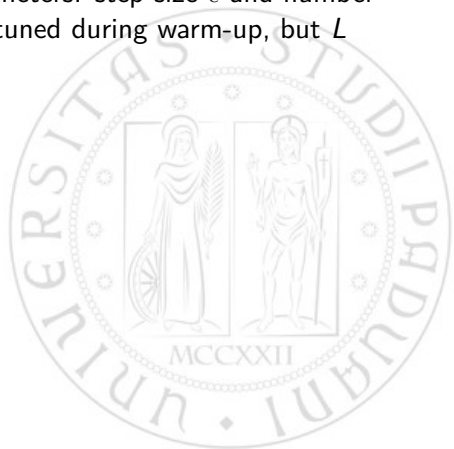
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Algorithm 1 Hamiltonian Monte Carlo

```
Given  $\theta^0, \epsilon, L, \mathcal{L}, M$ :  
for  $m = 1$  to  $M$  do  
  Sample  $r^0 \sim \mathcal{N}(0, I)$ .  
  Set  $\theta^m \leftarrow \theta^{m-1}, \tilde{\theta} \leftarrow \theta^{m-1}, \tilde{r} \leftarrow r^0$ .  
  for  $i = 1$  to  $L$  do  
    Set  $\tilde{\theta}, \tilde{r} \leftarrow \text{Leapfrog}(\tilde{\theta}, \tilde{r}, \epsilon)$ .  
  end for  
  With probability  $\alpha = \min \left\{ 1, \frac{\exp\{\mathcal{L}(\tilde{\theta}) - \frac{1}{2}\tilde{r} \cdot \tilde{r}\}}{\exp\{\mathcal{L}(\theta^{m-1}) - \frac{1}{2}r^0 \cdot r^0\}} \right\}$ , set  $\theta^m \leftarrow \tilde{\theta}, r^m \leftarrow -\tilde{r}$ .  
end for  
  
function Leapfrog( $\theta, r, \epsilon$ )  
  Set  $\tilde{r} \leftarrow r + (\epsilon/2)\nabla_{\theta}\mathcal{L}(\theta)$ .  
  Set  $\tilde{\theta} \leftarrow \theta + \epsilon\tilde{r}$ .  
  Set  $\tilde{r} \leftarrow \tilde{r} + (\epsilon/2)\nabla_{\theta}\mathcal{L}(\tilde{\theta})$ .  
  return  $\tilde{\theta}, \tilde{r}$ .
```

No U-Turn HMC

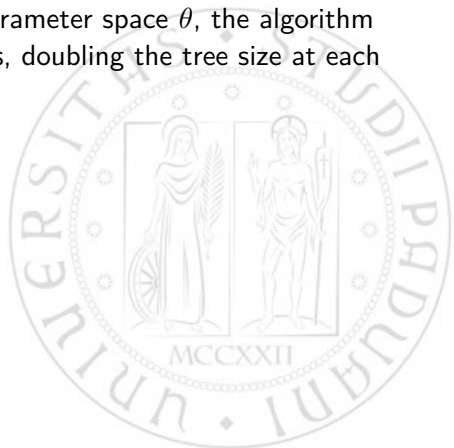
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- Starting from a point in the parameter space θ , the algorithm builds a set of candidate points, doubling the tree size at each iteration.
- To detect when to stop, the algorithm checks if the trajectory starts to turn back on itself: given a proposed point $\tilde{\theta}$, the algorithm checks if the dot product of the momentum vectors at θ and $\tilde{\theta}$ is negative.

$$(\theta - \tilde{\theta}) \cdot \frac{d}{dt}(\theta - \tilde{\theta}) = (\theta - \tilde{\theta}) \cdot \tilde{r} < 0$$

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- To preserve time reversibility, NUTS runs the Hamiltonian dynamics in both forward and backward directions.

Probability slicing

To simplify the derivation and implementation of the algorithm, NUTS uses a slice variable u with conditional distribution

$$p(u|\theta, r) = \mathcal{U}(u; [0, \exp\mathcal{L}(\theta) - \tfrac{1}{2}r \cdot r]).$$

This way $p(\theta, r|u) = \mathcal{U}(\theta, r; \theta', r' | \exp\mathcal{L}(\theta) - \tfrac{1}{2}r \cdot r \geq u)$.

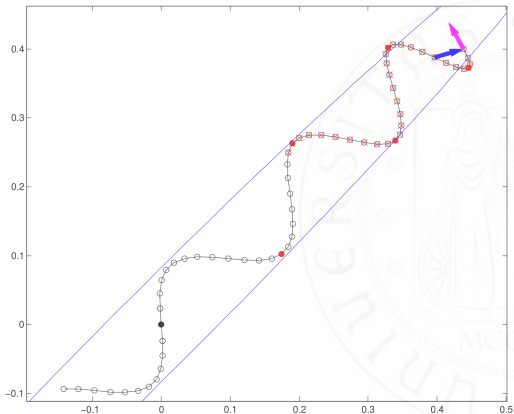


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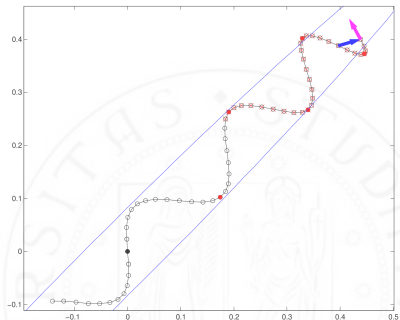
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High-level NUTS algorithm

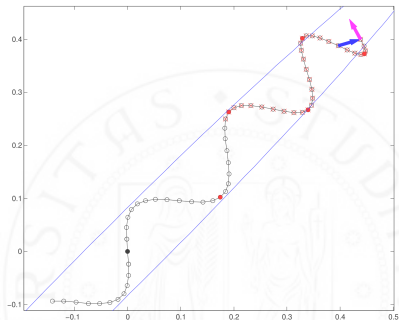
- The algorithm resamples $u|\theta, r$ at each iteration,



Example of a NUTS trajectory, with the slice variable u .

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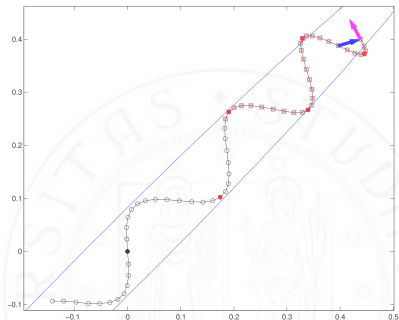
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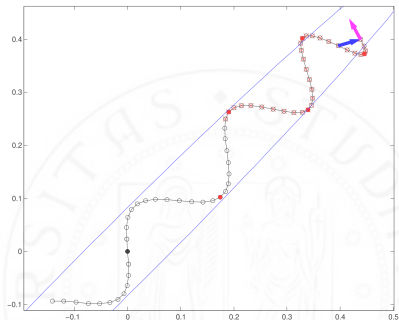
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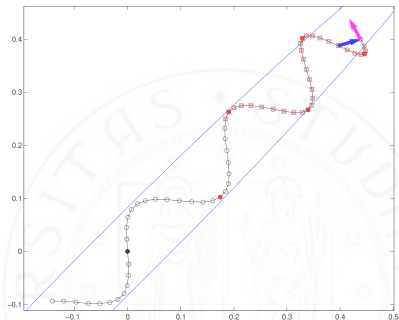
- The algorithm resamples $u|\theta, r$ at each iteration,
- uses Leapfrog integrator to trace a path forward and backwards first 1 step,
- then doubling for successive iterations.
- This doubling process implicitly builds a balanced binary tree whose leaf nodes correspond to position-momentum states.



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- uses Leapfrog integrator to trace a path forward and backwards first 1 step,
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- This doubling process implicitly builds a balanced binary tree whose leaf nodes correspond to position-momentum states.
- The doubling is halted when the subtrajectory starts to double back on itself.



Example of a NUTS trajectory, with the slice variable u .

Detailed NUTS algorithm

Given $\mathcal{B}$, set of position-momentum states traced by the leapfrog integrator, we pick deterministically $\mathcal{C}$, the set of candidate points.

Algorithm 2 Naive No-U-Turn Sampler

```
Given  $\theta^0, \epsilon, \mathcal{L}, M$ :  
for  $m = 1$  to  $M$  do  
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .  
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$   
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .  
  while  $s = 1$  do  
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .  
    if  $v_j = -1$  then  
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .  
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    end if  
    if  $s' = 1$  then  
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .  
    end if  
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .  
     $j \leftarrow j + 1$ .  
  end while  
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .  
end for  
  
function BuildTree( $\theta, r, u, v, j, \epsilon$ )  
  if  $j = 0$  then  
    Base case—take one leapfrog step in the direction  $v$ .  
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$ .  
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$   
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .  
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .  
  else  
    Recursion—build the left and right subtrees.  
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .  
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- The starting position-momentum state (θ, r) is added to $\mathcal{C}$.
- Each point in $\mathcal{C}$ must satisfy the slice variable condition $\exp \mathcal{L}(\theta) - \frac{1}{2} r \cdot r \geq u$.
- Each point in $\mathcal{C}$ must have the same probability of constructing the trajectory $\mathcal{B}$ and candidate set $\mathcal{C}$.

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Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
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  Initialize  $\theta^- = \theta^{m-1}, \theta^+ = \theta^{m-1}, r^- = r^0, r^+ = r^0, j = 0, \mathcal{C} = \{(\theta^{m-1}, r^0)\}, s = 1$ .
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    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if

```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

Algorithm 2 Naive No-U-Turn Sampler

```
Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

- Sample $r \sim \mathcal{N}(0, I)$,

Algorithm 2 Naive No-U-Turn Sampler

```
Given  $\theta^0, \epsilon, \mathcal{L}, M$ :  
for  $m = 1$  to  $M$  do  
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .  
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$   
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .  
  while  $s = 1$  do  
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .  
    if  $v_j = -1$  then  
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .  
    else  
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .  
    end if  
    if  $s' = 1$  then  
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .  
    end if  
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .  
     $j \leftarrow j + 1$ .  
  end while  
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .  
end for  
  
function BuildTree( $\theta, r, u, v, j, \epsilon$ )  
  if  $j = 0$  then  
    Base case—take one leapfrog step in the direction  $v$ .  
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .  
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$   
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .  
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .  
  else  
    Recursion—build the left and right subtrees.  
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$ .  
    if  $v = -1$  then  
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$ .  
    else  
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$ .  
    end if  
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .  
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .  
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .  
  end if
```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, \exp \mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}, \theta^+ = \theta^{m-1}, r^- = r^0, r^+ = r^0, j = 0, \mathcal{C} = \{(\theta^{m-1}, r^0)\}, s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, \exp \mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, \exp \mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.
- Sample $\theta^{t+1}, r \sim T(\theta^t, r, \mathcal{C})$, transition kernel.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}, \theta^+ = \theta^{m-1}, r^- = r^0, r^+ = r^0, j = 0, \mathcal{C} = \{(\theta^{m-1}, r^0)\}, s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t :

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, \exp \mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.
- Sample $\theta^{t+1}, r \sim T(\theta^t, r, \mathcal{C})$,
transition kernel.

Where the transition kernel $T(\theta, r, \mathcal{C})$ has leave the distribution over $\mathcal{C}$ invariant, valid because ϵ is uniform on the elements of $\mathcal{C}$ and $p(\theta, r | u, \mathcal{B}, \mathcal{C}, \epsilon) \propto \mathbb{I}[(\theta, r) \in \mathcal{C}]$.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

Algorithm 2 Naive No-U-Turn Sampler

```
Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

- First a single node for the starting state.

Algorithm 2 Naive No-U-Turn Sampler

```
Given  $\theta^0, \epsilon, \mathcal{L}, M$ :  
for  $m = 1$  to  $M$  do  
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .  
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$   
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .  
  while  $s = 1$  do  
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .  
    if  $v_j = -1$  then  
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .  
    else  
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .  
    end if  
    if  $s' = 1$  then  
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .  
    end if  
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .  
     $j \leftarrow j + 1$ .  
  end while  
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .  
end for  
  
function BuildTree( $\theta, r, u, v, j, \epsilon$ )  
  if  $j = 0$  then  
    Base case—take one leapfrog step in the direction  $v$ .  
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .  
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$   
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .  
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .  
  else  
    Recursion—build the left and right subtrees.  
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .  
    if  $v = -1$  then  
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .  
    else  
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .  
    end if  
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .  
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .  
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .  
  end if
```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

- First a single node for the starting state.
- Each iteration j proceeds with in a direction $v_j \sim \mathcal{U}(-1, 1)$ taking 2^j Leapfrog steps.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

- First a single node for the starting state.
- Each iteration j proceeds with in a direction $v_j \sim \mathcal{U}(-1, 1)$ taking 2^j Leapfrog steps.
- The iteration stops when the trajectory starts to double back on itself

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if

```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

- First a single node for the starting state.
- Each iteration j proceeds with in a direction $v_j \sim \mathcal{U}(-1, 1)$ taking 2^j Leapfrog steps.
- The iteration stops when the trajectory starts to double back on itself
- or when the error in the simulation becomes too large, $\mathcal{L}(\theta) - \frac{1}{2}r \cdot r - \log u < -\Delta_{\max}$.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end function

```

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

- First a single node for the starting state.
- Each iteration j proceeds with in a direction $v_j \sim \mathcal{U}(-1, 1)$ taking 2^j Leapfrog steps.
- The iteration stops when the trajectory starts to double back on itself
- or when the error in the simulation becomes too large, $\mathcal{L}(\theta) - \frac{1}{2}r \cdot r - \log u < -\Delta_{\max}$.

The doubling process defines the distribution $p(\mathcal{B}|\theta^t, r, u, \epsilon)$.

Algorithm 2 Naive No-U-Turn Sampler

```

Given  $\theta^0, \epsilon, \mathcal{L}, M$ :
for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if

```

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.

Algorithm 2 Naive No-U-Turn Sampler

Given $\theta^0, \epsilon, \mathcal{L}, M$:
for $m = 1$ to M **do**
 Resample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
 end if
 if $s' = 1$ **then**
 $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
 end if
 $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree($\theta, r, u, v, j, \epsilon$)
 if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v .
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
 $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
 $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$.
 return $\theta', r', \theta', r', \mathcal{C}', s'$.
 else
 Recursion—build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$.
 if $v = -1$ **then**
 $\theta^-, r^-, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$.
 end if
 $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
 return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
 end if

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.

Algorithm 2 Naive No-U-Turn Sampler

Given $\theta^0, \epsilon, \mathcal{L}, M$:

for $m = 1$ to M **do**

Resample $r^0 \sim \mathcal{N}(0, I)$.

Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$

Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.

while $s = 1$ **do**

Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.

if $v_j = -1$ **then**

$\theta^-, r^-, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.

else

$-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.

end if

for $s' = 1$ **then**

$\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.

end if

$s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.

$j \leftarrow j + 1$.

end while

Sample θ^m, r uniformly at random from $\mathcal{C}$.

end for

function BuildTree($\theta, r, u, v, j, \epsilon$)

if $j = 0$ **then**

Base case—take one leapfrog step in the direction v .

$\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.

$\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$

$s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$.

return $\theta', r', \theta', r', \mathcal{C}', s'$.

else

Recursion—build the left and right subtrees.

$\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.

if $v = -1$ **then**

$\theta^-, r^-, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.

else

$-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.

end if

$s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.

$\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.

return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.

end if

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.
- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.

Algorithm 2 Naive No-U-Turn Sampler

Given $\theta^0, \epsilon, \mathcal{L}, M$:

```

for  $m = 1$  to  $M$  do
  Resample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$ .
    end if
    if  $s' = 1$  then
       $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$ .
    end if
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
  Sample  $\theta^m, r$  uniformly at random from  $\mathcal{C}$ .
end for

function BuildTree( $\theta, r, u, v, j, \epsilon$ )
  if  $j = 0$  then
    Base case—take one leapfrog step in the direction  $v$ .
     $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$ .
     $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$ 
     $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$ .
    return  $\theta', r', \theta', r', \mathcal{C}', s'$ .
  else
    Recursion—build the left and right subtrees.
     $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon)$ .
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon)$ .
    else
       $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon)$ .
    end if
     $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$ .
    return  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$ .
  end if
end if

```

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.
- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.
- If a global U turn condition is met, the doubling process is terminated.

Algorithm 2 Naive No-U-Turn Sampler

Given $\theta^0, \epsilon, \mathcal{L}, M$:
for $m = 1$ to M **do**
 Resample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}(\{0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}\})$.
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
 end if
 if $s' = 1$ **then**
 $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
 end if
 $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree($\theta, r, u, v, j, \epsilon$)
if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v .
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$.
 $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
 $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$.
 return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
 Recursion—build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
 if $v = -1$ **then**
 $\theta^-, r^-, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
 end if
 $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
 return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.
- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.
- If a global U turn condition is met, the doubling process is terminated.
- The new position-momentum state (θ^{t+1}, r) is sampled from the set of candidate points $\mathcal{C}$.

Algorithm 2 Naive No-U-Turn Sampler

Given $\theta^0, \epsilon, \mathcal{L}, M$:
for $m = 1$ to M **do**
 Resample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}(\{0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}\})$.
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
 end if
 if $s' = 1$ **then**
 $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
 end if
 $s \leftarrow s \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree($\theta, r, u, v, j, \epsilon$)
if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v .
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v, \epsilon)$.
 $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
 $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' > \log u - \Delta_{\max}]$.
 return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
 Recursion—build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
 if $v = -1$ **then**
 $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
 else
 $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
 end if
 $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
 return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

Algorithm 6 No-U-Turn Sampler with Dual Averaging

```

Given  $\theta^0, \delta, \mathcal{L}, M, M^{\text{adapt}}$ ;
Set  $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$ ,  $\mu = \log(10\epsilon_0)$ ,  $\bar{\epsilon}_0 = 1$ ,  $\bar{H}_0 = 0$ ,  $\gamma = 0.05$ ,  $t_0 = 10$ ,  $\kappa = 0.75$ .
for  $m = 1$  to  $M$  do
  Sample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}$ ,  $\theta^+ = \theta^{m-1}$ ,  $r^- = r^0$ ,  $r^+ = r^0$ ,  $j = 0$ ,  $\theta^m = \theta^{m-1}$ ,  $n = 1$ ,  $s = 1$ .
  while  $s = 1$  do
    Choose a direction  $v_j \sim \text{Uniform}(\{-1, 1\})$ .
    if  $v_j = -1$  then
       $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$ .
    else
       $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$ .
    end if
    if  $s' = 1$  then
      With probability  $\min\{1, \frac{n'}{n}\}$ , set  $\theta^m \leftarrow \theta'$ .
    end if
     $n \leftarrow n + n'$ .
     $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ .
     $j \leftarrow j + 1$ .
  end while
if  $m \leq M^{\text{adapt}}$  then
  Set  $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right) \bar{H}_{m-1} + \frac{1}{m+t_0} \left(\delta - \frac{\alpha}{n_\alpha}\right)$ .
  Set  $\log \epsilon_m = \mu - \frac{\gamma m}{2\bar{H}_m}$ ,  $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \epsilon_{m-1}$ .
  else
    Set  $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$ .
  end if
end for

function  $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$ 
if  $j = 0$  then
  Base case—take one leapfrog step in the direction  $v$ .
   $\theta', r' \leftarrow \text{Leapfrog}(\theta, v)$ .
   $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$ .
   $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$ .
  return  $\theta', r', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2}r^0 \cdot r^0\}\}, 1$ .
else
  Recursion—implicitly build the left and right subtrees.
   $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon, \theta^0, r^0)$ .
  if  $s' = 1$  then
    if  $v = -1$  then
       $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon, \theta^0, r^0)$ .
    else
       $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon, \theta^0, r^0)$ .
    end if
    if
      With probability  $\frac{n''}{n' + n''}$ , set  $\theta' \leftarrow \theta''$ .
     $\text{Set } \alpha' \leftarrow \alpha' + \alpha'', n'_\alpha \leftarrow n'_\alpha + n''_\alpha$ .
     $s \leftarrow s' \mathbb{I}[(\theta^- - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$ 
     $n' \leftarrow n' + n''$ 
  end if
  return  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$ .
end if

```

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given $\theta^0, \delta, \mathcal{L}, M, M^{\text{adapt}}$;
 Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
 Sample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$.
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\theta^m = \theta^{m-1}$, $n = 1$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
 end if
 if $s' = 1$ **then**
 With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
 end if
 $n \leftarrow n + n'$.
 $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 if $m \leq M^{\text{adapt}}$ **then**
 Set $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right) \bar{H}_{m-1} + \frac{1}{m+t_0} (\delta - \frac{\alpha}{n_\alpha})$.
 Set $\log \epsilon_m = \mu - \frac{\gamma \bar{m}}{2} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \epsilon_{m-1}$.
 else
 Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
 end if
 end for
function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v .
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v)$.
 $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\text{max}} + \mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2}r^0 \cdot r^0\}\}, 1$.
else
 Recursion—implicitly build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon, \theta^0, r^0)$.
 if $s' = 1$ **then**
 if $v = -1$ **then**
 $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon, \theta^0, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon, \theta^0, r^0)$.
 end if
 With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
 Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
 $s' \leftarrow s' \mathbb{I}[(\theta^- - \theta^+) \cdot r^- \geq 0] \mathbb{I}[(\theta^- - \theta^+) \cdot r^+ \geq 0]$.
 $n' \leftarrow n' + n''$.
 end if
 return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.
- Single Proposed State kept in memory, by sampling a candidate for each subtree and drawing from them using their relative weights (i.e. the size of the subtree).

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given $\theta^0, \delta, \mathcal{L}, M, M^{\text{adapt}}$;
 Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ **to** M **do**
 Sample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$.
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\theta^m = \theta^{m-1}$, $n = 1$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
 end if
 if $s' = 1$ **then**
 With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
 end if
 $n \leftarrow n + n'$.
 $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 if $m \leq M^{\text{adapt}}$ **then**
 Set $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right) \bar{H}_{m-1} + \frac{1}{m+t_0} (\delta - \frac{\alpha}{n_\alpha})$.
 Set $\log \bar{\epsilon}_m = \mu - \frac{\gamma}{2\bar{H}_m} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \bar{\epsilon}_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
 else
 Set $\bar{\epsilon}_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
 end if
 end for
function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v.
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v)$.
 $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 $s' \leftarrow \mathbb{I}[u \leq \exp\{\Delta_{\text{max}} + \mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2}r^0 \cdot r^0\}\}, 1$.
else
 Recursion—implicitly build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon, \theta^0, r^0)$.
 if $s' = 1$ **then**
 if $v = -1$ **then**
 $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon, \theta^0, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon, \theta^0, r^0)$.
 end if
 With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
 Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
 $s' \leftarrow s' \mathbb{I}[(\theta^- - \theta^+) \cdot r^- \geq 0] \mathbb{I}[(\theta^- - \theta^+) \cdot r^+ \geq 0]$.
 $n' \leftarrow n' + n''$.
 end if
 return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end function

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.
- Single Proposed State kept in memory, by sampling a candidate for each subtree and drawing from them using their relative weights (i.e. the size of the subtree).
- Adaptive tuning of the step size parameter ϵ , using dual averaging.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given $\theta^0, \delta, \mathcal{L}, M, M^{\text{adapt}}$;
 Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
 Sample $r^0 \sim \mathcal{N}(0, I)$.
 Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0 \cdot r^0)\}])$.
 Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\theta^m = \theta^{m-1}$, $n = 1$, $s = 1$.
 while $s = 1$ **do**
 Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
 if $v_j = -1$ **then**
 $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
 end if
 if $s' = 1$ **then**
 With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
 end if
 $n \leftarrow n + n'$.
 $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
 $j \leftarrow j + 1$.
 end while
 if $m \leq M^{\text{adapt}}$ **then**
 Set $\bar{H}_m = \left(1 - \frac{1}{m+t_0}\right) \bar{H}_{m-1} + \frac{1}{m+t_0} (\delta - \frac{\alpha}{n_\alpha})$.
 Set $\log \bar{\epsilon}_m = \mu - \frac{\gamma}{2} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \bar{\epsilon}_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
 else
 Set $\bar{\epsilon}_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
 end if
 end for

function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
 Base case—take one leapfrog step in the direction v.
 $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v)$.
 $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 $s' \leftarrow \mathbb{I}[u \leq \exp\{\Delta_{\text{max}} + \mathcal{L}(\theta') - \frac{1}{2}r' \cdot r'\}]$.
 return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2}r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2}r^0 \cdot r^0\}\}, 1$.
else
 Recursion—implicitly build the left and right subtrees.
 $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j-1, \epsilon, \theta^0, r^0)$.
 if $s' = 1$ **then**
 if $v = -1$ **then**
 $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j-1, \epsilon, \theta^0, r^0)$.
 else
 $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j-1, \epsilon, \theta^0, r^0)$.
 end if
 With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
 Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
 $s' \leftarrow s' \mathbb{I}[(\theta^- - \theta^+) \cdot r^- \geq 0] \mathbb{I}[(\theta^- - \theta^+) \cdot r^+ \geq 0]$
 $n' \leftarrow n' + n''$.
 end if
 return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Dual Averaging for ϵ

ϵ is set using stochastic optimization with vanishing adaptation.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

```

Given  $\theta^0, \delta, \mathcal{L}, M, M^{\text{adapt}}$ ;
Set  $\epsilon_0 = \text{FindReasonableEpsilon}(\theta), \mu = \log(10\epsilon_0), \bar{\epsilon}_0 = 1, \bar{R}_0 = 0, \gamma = 0.05, t_0 = 10, \kappa = 0.75$ .
for  $m = 1$  to  $M$  do
  Sample  $r^0 \sim \mathcal{N}(0, I)$ .
  Resample  $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2}r^0, r^0)\}])$ 
  Initialize  $\theta^- = \theta^{m-1}, \theta^+ = \theta^{m-1}, r^- = r^0, r^+ = r^0, j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$ .
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        if  $s' = 1$  then
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$$x_{t+1} \leftarrow \mu - \frac{\sqrt{t}}{\gamma} \frac{1}{t+t_0} \sum_{i=1}^t H_i,$$

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with $\eta_t = t^{-k}$.

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$$\begin{aligned} x_{t+1} &\leftarrow \mu - \frac{\sqrt{t}}{\gamma} \frac{1}{t+t_0} \sum_{i=1}^t H_i, \\ \bar{x}_{t+1} &\leftarrow \bar{x}_t + \eta_t x_{t+1} + (1 - \eta_t) \bar{x}_t \\ &\text{with } \eta_t = t^{-k}. \end{aligned}$$

$\mu = \log 10\epsilon_1$ is a freely chosen point, $\gamma = 0.05$ is a free parameter for the speed of convergence, and $t_0 = 10$ is a free parameter that stabilizes the initial iterations and $k = 0.75 \in (0.5, 1]$.

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Heuristic function for ϵ

Convergence of Dual Averaging for ϵ is faster starting from a reasonable setting of parameters.

Algorithm 4 Heuristic for choosing an initial value of ϵ

function FindReasonableEpsilon(θ)

Initialize $\epsilon = 1$, $r \sim \mathcal{N}(0, I)$. (I denotes the identity matrix.)

Set $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, \epsilon)$.

$a \leftarrow 2\mathbb{I}\left[\frac{p(\theta', r')}{p(\theta, r)} > 0.5\right] - 1$.

while $\left(\frac{p(\theta', r')}{p(\theta, r)}\right)^a > 2^{-a}$ **do**

$\epsilon \leftarrow 2^a \epsilon$.

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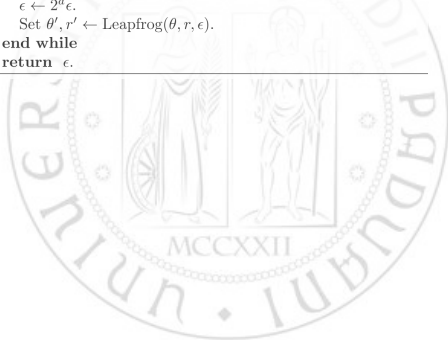
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Now the algorithm only requires an acceptance probability δ as input, and a number of iterations M^{adapt} for the adaptation phase.

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In order to implement the algorithm we need some tools:

- Automatic differentiation
- Vectorial computation for computations in high dimension space
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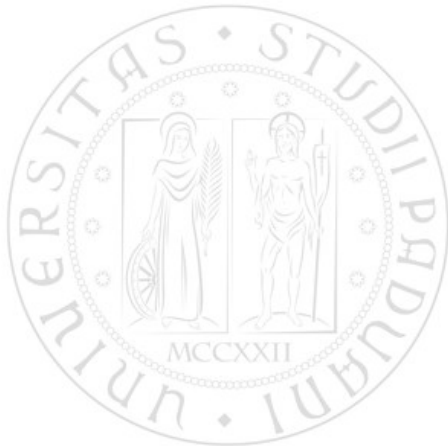
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JAX (Iterative & Functional)

- Recursion not supported by XLA.
- Functional loops (e.g., `lax.fori_loop`).
- Pure functions: no side effects.
- Immutable values.
- Static control flow for compilation.
- Works with `jit`, `vmap`, `grad`.

Recurion Vs Iteration

Algorithm 1: recursion

Input: n

Output: $n!$

if $n \leq 1$ **then**

return 1

else

return $n \cdot \text{factorial}(n - 1)$

Algorithm 2: iteration

Input: n

Output: $n!$

$acc \leftarrow 1$;

for $i \leftarrow 1$ **to** n **do**

$acc \leftarrow acc \cdot i$;

return acc

Code implementation

```
1 # Python factorial
2 def factorial_recursive(n: int) -> int:
3     if n <= 1:
4         return 1
5     return n * factorial_recursive(n - 1)
```



Code implementation

```
1 # Python factorial
2 def factorial_recursive(n: int) -> int:
3     if n <= 1:
4         return 1
5     return n * factorial_recursive(n - 1)
```

```
1 # JAX factorial
2 import jax
3 import jax.numpy as jnp
4 from jax import lax
5
6 def factorial_jax(n: int):
7     def body(i, acc):
8         return acc * (i + 1)
9     return lax.fori_loop(0, n, body, 1)
```

NUTS Architecture Overview — Part I

1. Base Components

`log_p`: Joint log-density $L(\theta) - \frac{1}{2}r^T r$. `stop_criterion`: Detects U-turns to stop tree expansion. `Leapfrog`: Hamiltonian integrator (reversible, volume-preserving).

2. Tree Construction

`build_tree_base`: Depth-0 subtree via one leapfrog step. `BuildTree`: Recursive tree expansion using `lax.fori_loop`; aggregates valid states, acceptance stats, and U-turn checks.

3. NUTS Kernel

`NUTS_one_step`: Full NUTS transition (Algorithm 3). Samples momentum, performs slice sampling, builds the tree, selects candidate state, computes acceptance rate.

4. Step Size Initialization & Adaptation

`FindReasonableEpsilon`: Heuristic search for initial step size.

`dual_avg_init`: Initializes dual-averaging parameters.

`dual_avg_update`: Adapts step size toward target acceptance.

5. Warmup Phase

`NUTS_warmup`: Runs warmup (Algorithm 6), updating dual averaging and returning tuned step size ϵ_{final} . `WarmupConfig`: Stores warmup settings.

6. Full Sampler

`nuts_sampler`: NUTS without warmup. `NUTS_sampler_wu`: Full sampler with warmup + sampling phase.